

LOAN APPROVAL PREDICTION USING DECISION TREE

Internship Mini Project Report

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Internship: Machine Learning Internship at Geonixa

Technology Used: Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-Learn, Streamlit, Jupyter Notebook

1. Introduction

The Loan Approval Prediction System is a Machine Learning-based mini project developed to predict whether a customer's loan application is likely to be approved or rejected.

The system uses the **Decision Tree Classification Algorithm** and analyzes various applicant details such as:

- Annual Income
- Credit Score
- Loan Amount
- Employment Status
- Education Level
- Existing Debt

The project was developed as part of the Machine Learning Internship program to gain practical experience in machine learning, data preprocessing, data visualization, model development, and deployment using Streamlit.

2. Objectives

- To understand Machine Learning concepts
- To perform data preprocessing using Python
- To apply Decision Tree Classification
- To evaluate model performance
- To visualize data using graphs
- To build an interactive Streamlit application

- To develop a real-world machine learning project

3. Features of the Project

- Dataset Loading
- Data Preprocessing
- Label Encoding
- Feature Selection
- Decision Tree Model Training
- Model Evaluation
- Confusion Matrix Generation
- Feature Importance Analysis
- Correlation Heatmap
- Streamlit Web Application
- Real-Time Loan Prediction

4. Technologies Used

Technology	Purpose
Python	Main Programming Language
Pandas	Data Manipulation
NumPy	Numerical Operations
Matplotlib	Data Visualization
Seaborn	Statistical Visualization
Scikit-Learn	Machine Learning Model
Streamlit	Web Application Development
Jupyter Notebook	Development Environment

5. Project Explanation

The project predicts loan approval status using applicant information.

The dataset contains:

200 Rows

7 Columns

The following workflow was implemented:

Step 1: Data Loading

The dataset was loaded using Pandas.

Step 2: Data Preprocessing

Categorical features were converted into numerical values using Label Encoding.

Step 3: Model Training

A Decision Tree Classifier was trained using the processed dataset.

Step 4: Model Evaluation

The model performance was evaluated using:

- Accuracy Score
- Classification Report
- Confusion Matrix

Step 5: Streamlit Application

A user-friendly web application was developed to allow users to enter applicant details and receive instant loan approval predictions.

6. Dataset Information

Dataset Shape:

(200, 7)

Features:

- Annual_Income
- Credit_Score
- Loan_Amount
- Employment_Status
- Education_Level
- Existing_Debt
- Loan_Status

Target Variable:

Loan_Status

7. Model Performance

Algorithm Used:

Decision Tree Classifier

Accuracy Achieved:

97.5%

The model successfully classified loan applications as Approved or Rejected with high accuracy.

8. Sample Prediction Output

Input

Annual Income : 900000

Credit Score : 850

Loan Amount : 100000

Employment Status : Employed

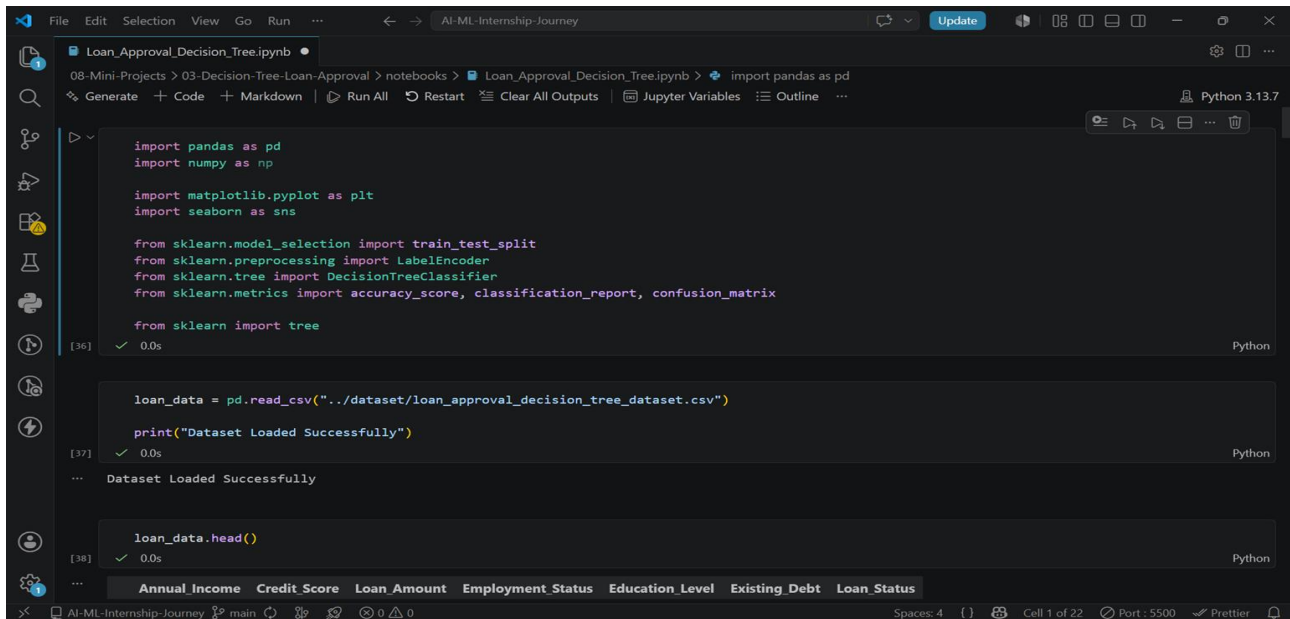
Education Level : Graduate

Existing Debt : 10000

Output

✓Loan Application Status: Approved

9. Python Source Code and Output Screenshots



```
import pandas as pd
import numpy as np

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix

from sklearn import tree
```

[36] ✓ 0.0s Python

```
loan_data = pd.read_csv("../dataset/loan_approval_decision_tree_dataset.csv")

print("Dataset Loaded Successfully")
```

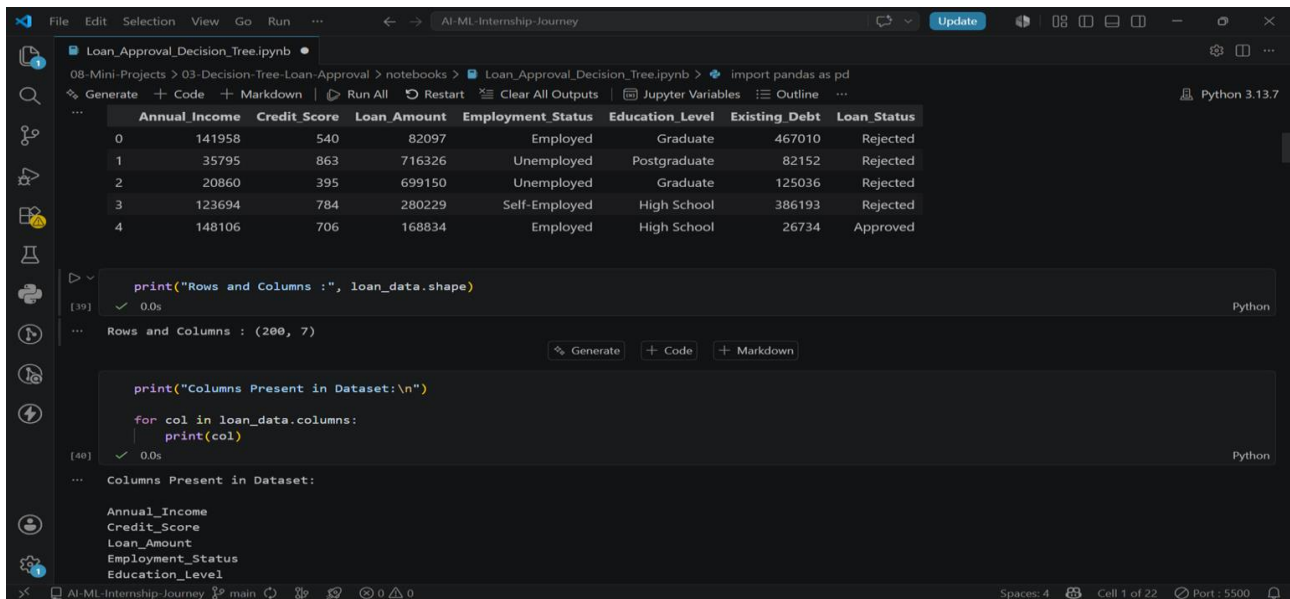
[37] ✓ 0.0s Python

... Dataset Loaded Successfully

```
loan_data.head()
```

[38] ✓ 0.0s Python

	Annual_Income	Credit_Score	Loan_Amount	Employment_Status	Education_Level	Existing_Debt	Loan_Status
0	141958	540	82097	Employed	Graduate	467010	Rejected
1	35795	863	716326	Unemployed	Postgraduate	82152	Rejected
2	20860	395	699150	Unemployed	Graduate	125036	Rejected
3	123694	784	280229	Self-Employed	High School	386193	Rejected
4	148106	706	168834	Employed	High School	26734	Approved



```
print("Rows and Columns :", loan_data.shape)
```

[39] ✓ 0.0s Python

... Rows and Columns : (200, 7)

```
print("Columns Present in Dataset:\n")

for col in loan_data.columns:
    print(col)
```

[40] ✓ 0.0s Python

... Columns Present in Dataset:

Annual_Income
Credit_Score
Loan_Amount
Employment_Status
Education_Level

```
File Edit Selection View Go Run ... AI-ML-Internship-Journey Update
Loan_Approval_Decision_Tree.ipynb
08-Mini-Projects > 03-Decision-Tree-Loan-Approval > notebooks > Loan_Approval_Decision_Tree.ipynb > import pandas as pd
Generate + Code + Markdown Run All Restart Clear All Outputs Jupyter Variables Outline Python 3.13.7

... Columns Present in Dataset:

Annual_Income
Credit_Score
Loan_Amount
Employment_Status
Education_Level
Existing_Debt
Loan_Status

loan_data.info()
[41] ✓ 0.0s Python

... <class 'pandas.DataFrame'>
RangeIndex: 200 entries, 0 to 199
Data columns (total 7 columns):
# Column Non-Null Count Dtype
---
0 Annual_Income 200 non-null int64
1 Credit_Score 200 non-null int64
2 Loan_Amount 200 non-null int64
3 Employment_Status 200 non-null str
4 Education_Level 200 non-null str
5 Existing_Debt 200 non-null int64
6 Loan_Status 200 non-null str
dtypes: int64(4), str(3)
memory usage: 16.6 KB
```

```
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08-Mini-Projects > 03-Decision-Tree-Loan-Approval > notebooks > Loan_Approval_Decision_Tree.ipynb > import pandas as pd
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print("Missing Values:\n")
loan_data.isnull().sum()
[42] ✓ 0.0s Python

... Missing Values:

... Annual_Income 0
Credit_Score 0
Loan_Amount 0
Employment_Status 0
Education_Level 0
Existing_Debt 0
Loan_Status 0
dtype: int64

loan_data.describe()
[43] ✓ 0.0s Python

... Annual_Income Credit Score Loan_Amount Existing Debt
count 200.000000 200.000000 200.000000 200.000000
mean 89085.090000 594.375000 497014.710000 244203.480000
std 39393.992473 166.619732 280567.876725 148386.550867
min 20206.000000 301.000000 53051.000000 846.000000
25% 57161.750000 453.500000 240775.750000 115461.000000
```

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Loan_Approval_Decision_Tree.ipynb

08-Mini-Projects > 03-Decision-Tree-Loan-Approval > notebooks > Loan_Approval_Decision_Tree.ipynb > import pandas as pd

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25%	57161.750000	453.500000	240775.750000	115461.000000
50%	89285.500000	582.000000	505286.500000	246257.500000
75%	124547.000000	738.750000	740729.000000	383915.000000
max	149981.000000	886.000000	999597.000000	499097.000000

```
encoder = LabelEncoder()

categorical_columns = [
    "Employment_Status",
    "Education_Level",
    "Loan_Status"
]

for column in categorical_columns:
    loan_data[column] = encoder.fit_transform(loan_data[column])

loan_data.head()
```

[44] ✓ 0.0s Python

	Annual_Income	Credit_Score	Loan_Amount	Employment_Status	Education_Level	Existing_Debt	Loan_Status
0	141958	540	82097	0	0	467010	1
1	35795	863	716326	2	2	82152	1
2	20860	395	699150	2	0	125036	1
3	123694	784	280229	1	1	386193	1
4	148106	706	168834	0	1	26734	0

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Loan_Approval_Decision_Tree.ipynb

08-Mini-Projects > 03-Decision-Tree-Loan-Approval > notebooks > Loan_Approval_Decision_Tree.ipynb > import pandas as pd

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```
X = loan_data.drop("Loan_Status", axis=1)

y = loan_data["Loan_Status"]

print("Features Shape :", X.shape)
print("Target Shape :", y.shape)
```

[45] ✓ 0.0s Python

Features Shape : (200, 6)
Target Shape : (200,)

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("Training Data :", X_train.shape)
print("Testing Data :", X_test.shape)
```

[46] ✓ 0.0s Python

Training Data : (160, 6)
Testing Data : (40, 6)

```
loan_model = DecisionTreeClassifier()
```

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```
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loan_model = DecisionTreeClassifier(
    criterion="gini",
    random_state=42
)

loan_model.fit(X_train, y_train)

print("Model Trained Successfully")

[47] ✓ 0.0s Python
... Model Trained Successfully

predictions = loan_model.predict(X_test)

predictions

[48] ✓ 0.0s Python
... array([0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1,
        1, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 0, 1, 1, 1, 1, 0])

accuracy = accuracy_score(y_test, predictions)

print("Accuracy :", accuracy)

[49] ✓ 0.0s Python
... Accuracy : 0.975

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```

```
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print(classification_report(y_test, predictions))

[50] ✓ 0.0s Python
...
      precision    recall  f1-score   support

     0       1.00      0.88      0.93         8
     1       0.97      1.00      0.98        32

 accuracy          0.97         40
  macro avg          0.98         40
 weighted avg          0.97         40

matrix = confusion_matrix(y_test, predictions)

print(matrix)

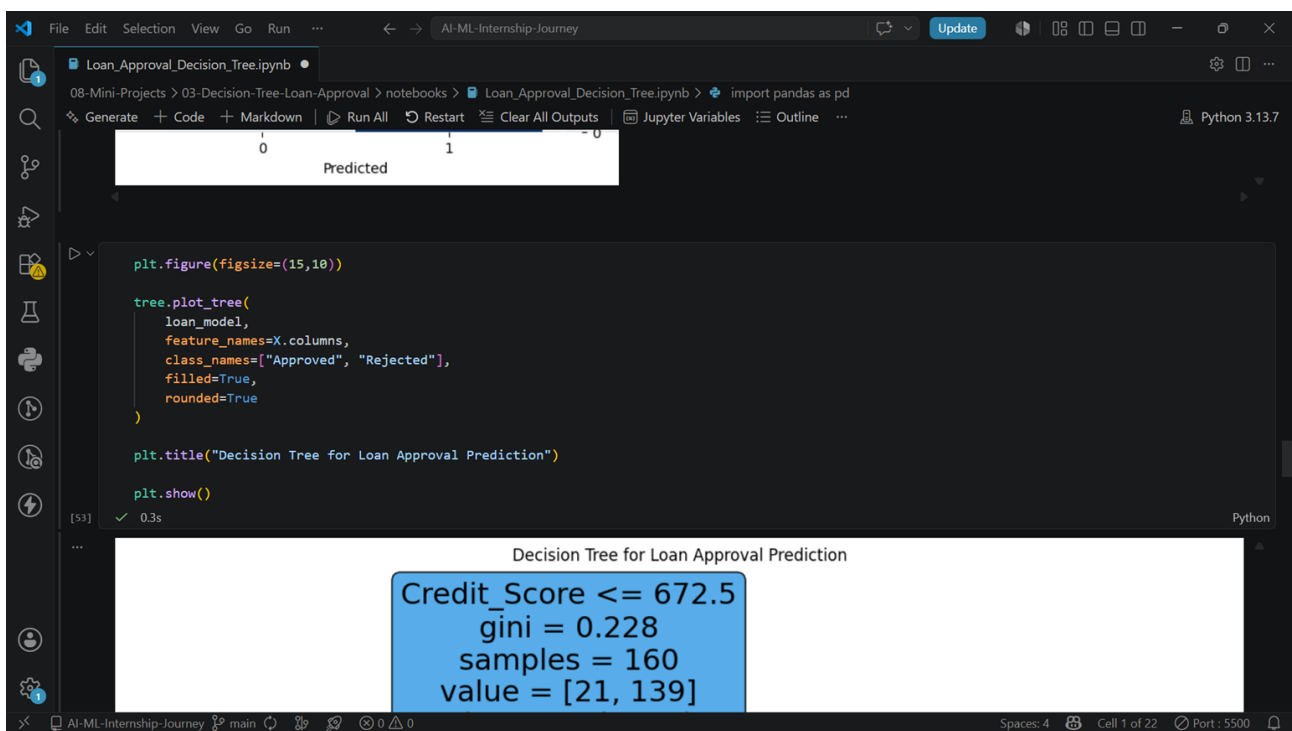
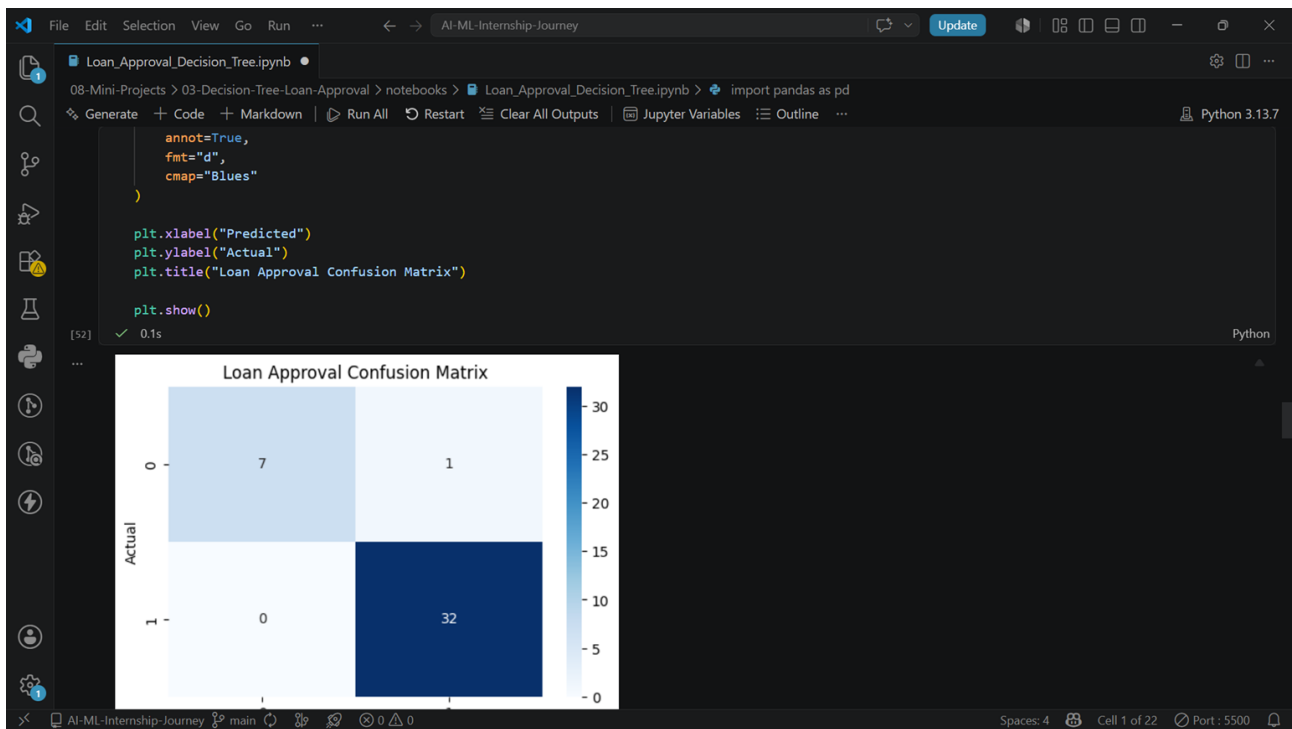
[51] ✓ 0.0s Python
... [[ 7  1]
     [ 0 32]]

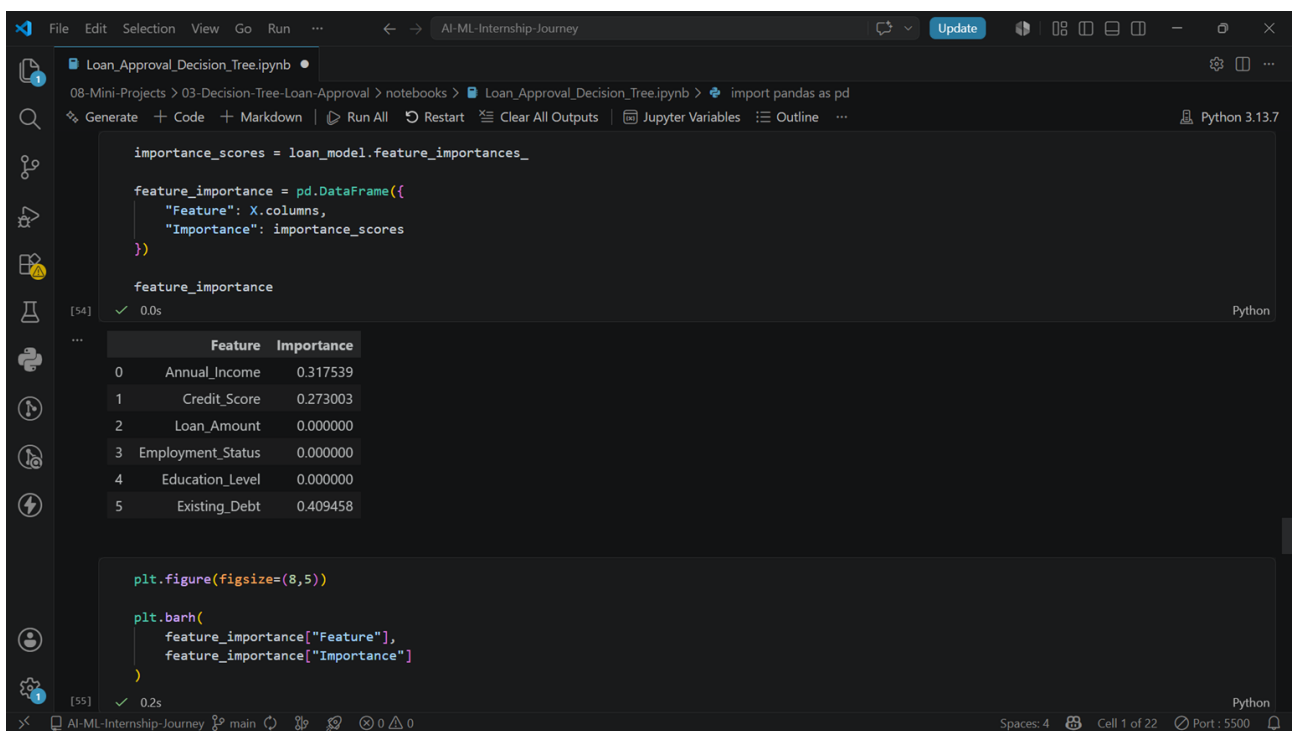
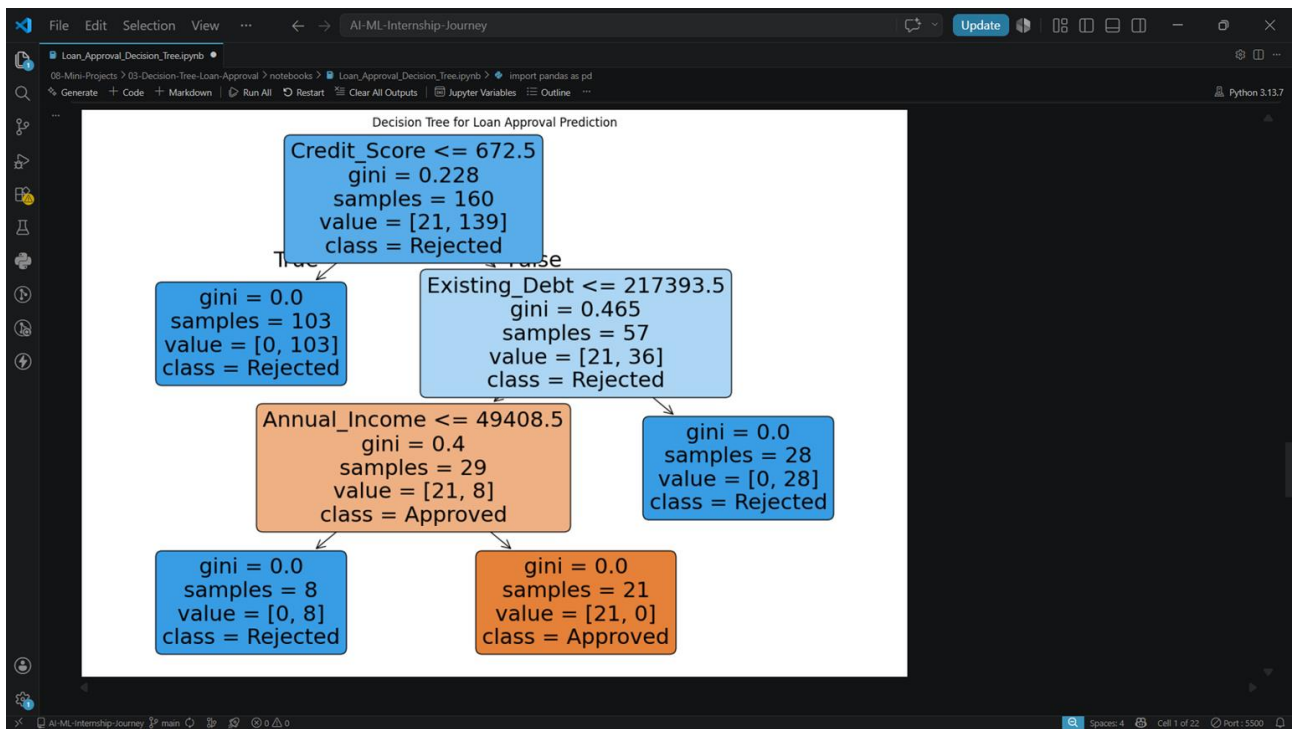
plt.figure(figsize=(6,4))

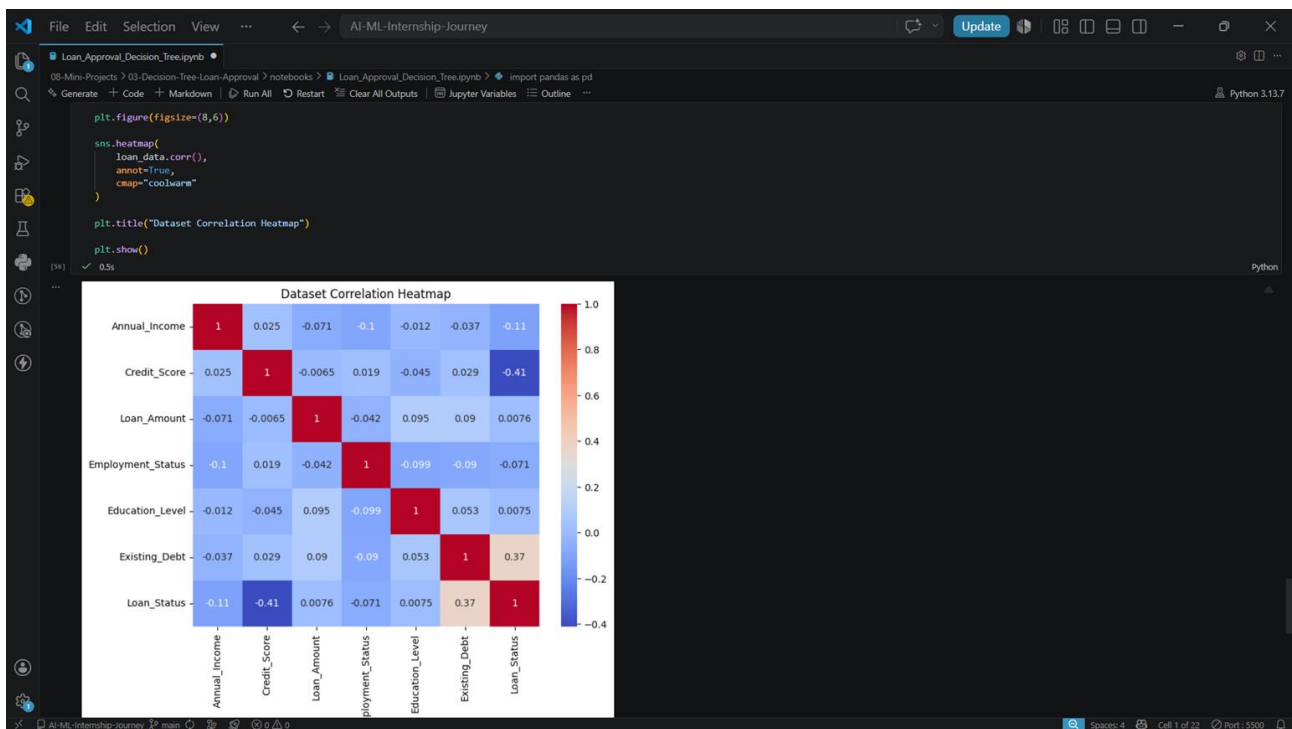
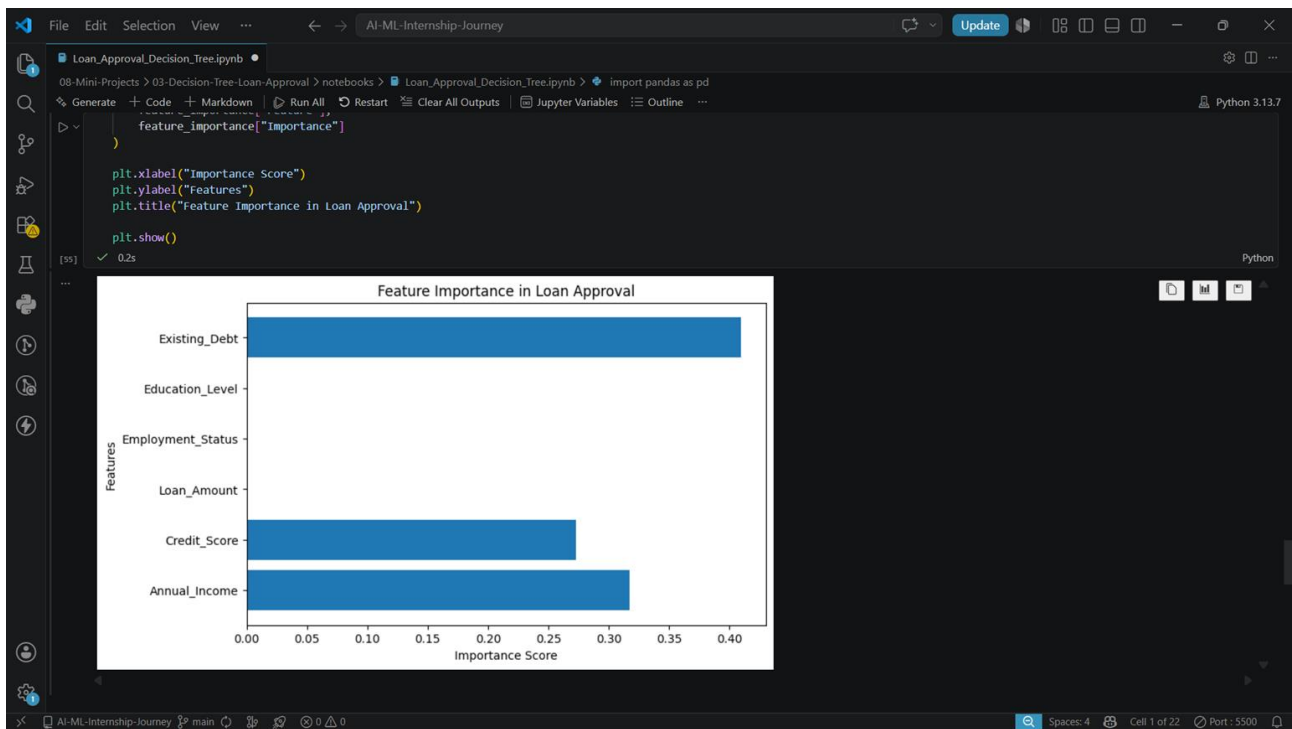
sns.heatmap(
    matrix,

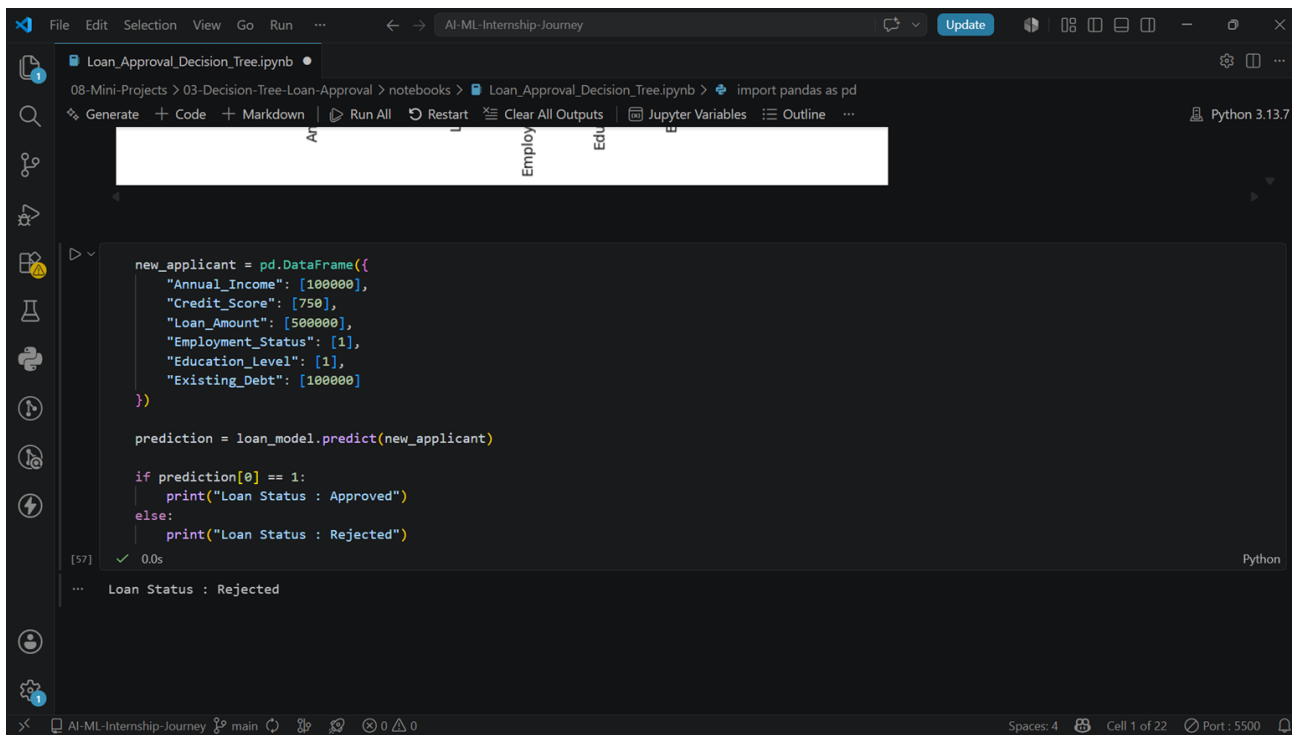
[52] ✓ 0.1s Python

AI-ML-Internship-Journey main 0 0 0 Spaces: 4 Cell 1 of 22 Port: 5500
```







```
new_applicant = pd.DataFrame({
    "Annual_Income": [100000],
    "Credit_Score": [750],
    "Loan_Amount": [500000],
    "Employment_Status": [1],
    "Education_Level": [1],
    "Existing_Debt": [100000]
})

prediction = loan_model.predict(new_applicant)

if prediction[0] == 1:
    print("Loan Status : Approved")
else:
    print("Loan Status : Rejected")
```

[57] ✓ 0.0s Python

... Loan Status : Rejected

11. Learning Outcomes

Through this project, the following concepts were learned:

- Data Preprocessing
- Label Encoding
- Classification Algorithms
- Decision Tree Modeling
- Model Evaluation Techniques
- Data Visualization
- Streamlit Development
- Machine Learning Workflow
- Real-Time Prediction Systems

12. Conclusion

The Loan Approval Prediction System was successfully developed using the Decision Tree Machine Learning Algorithm.

The model achieved an accuracy of **97.5%** and was successfully integrated into a Streamlit web application for real-time loan approval prediction.

The project provided practical experience in machine learning model development, evaluation, visualization, and deployment.

13. References

- [Python Official Documentation](#)
- [Scikit-Learn Documentation](#)
- [Pandas Documentation](#)
- [Streamlit Documentation](#)
- [Geonixa Internship Learning Materials](#)